QUALITY ASSURANCE · TRAINER SCRIPT

QA Inspector Training Script

The trainer's companion to the UQMES QA walkthrough

System	UQMES — Demo Company tenant
Teaches against	WO-QA-INSPECT-01 · Common Rail Injector · 8 parts
Audience	New QA inspectors and their leads
Runtime	~50 min · journeys 1–11, manager block optional

Contents

Audience: New QA inspectors (and their leads) learning to use UQMES on the shop floor. **Not a sales demo** — this teaches the software, not the pitch. Assume the trainee knows *inspection*; they need to learn *this tool*. **Runtime:** ~50 min for the inspector journeys (1–11); the manager gates and authoring are a separate, optional block. **Format:** classroom or 1:1 at a workstation, with a printed traveler in hand and a barcode reader if you have one. **Style:** click-runbook. Each journey tells you which button to press and where you'll land. It is the **trainer's companion to `UQMES_ONBOARDING_WALKTHROUGH.md`** — same demo work order, same exhibits; the walkthrough is the self-serve read, this is the taught version. Section cross-references (§N) point into the walkthrough.

> **Before class:** run `python manage.py seed_demo` and log in fresh once — the seed invalidates open sessions. All serials and record numbers are deterministic per reseed, *except* the auto-generated `DISP-2026-####` / `QR-2026-####` / `OSP-2026-####` numbers, which rotate — teach to the **part and journey**, never to a specific auto-number.

> **Screenshots:** this script was rebuilt onto the WO-QA-INSPECT-01 scenario and its screenshots have **not been recaptured yet** — see *Refreshing screenshots* at the bottom. Until then, teach from the live screen, not from images.

Trainer preparation checklist

Do these ~15 min before the trainee arrives. None survive being done live.

- ☐ `python manage.py seed_demo` on the training instance. Wait for a clean finish, no red errors.
- ☐ Log in fresh as **`sarah.qa@demo.ambac.com`** (`demo123`). Confirm the home page shows the **FPI banner** (two "First piece waiting" rows — `WO-2024-0048-A` and `WO-QA-INSPECT-01 · INJ-QA-INSPECT-001`) and an Inbox with rows. If the FPI banner is missing, re-seed.
- ☐ Have **Maria's** login ready — `maria.qa@demo.ambac.com` (`demo123`). You need her to co-sign the disposition decision in Journey 6; she's the QA Manager. Journey 6 does not complete on Sarah's login alone.
- ☐ Print the **Traveler** for **WO-QA-INSPECT-01**: WO Detail (`/workorder/$workOrderId`) → **Traveler** button. Verify the header barcode scans. (A pre-generated copy lives at `WO-QA-INSPECT-01_traveler.pdf`.)
- ☐ Spare barcode reader if you have one — a class stalled on hardware loses attention fast.
- ☐ Optional: a second browser/incognito window logged in as an operator (`mike.ops@demo.ambac.com`, `demo123`) to show the operator/inspector hand-off.
- ☐ Spot-check three paths on the training instance so you adjust expectations *before* class, not during:
  - Journey 4 (FPI): part **001**'s runtime shows "2 of 2 confirmed" and the **Sign off & pass / Fail / Waive** banner.
  - Journey 6 (fail → disposition): entering an out-of-spec flow value on **003** derives a FAIL and quarantines it; opening its disposition and setting a type as **Sarah** raises the **co-sign** dialog. **This is**

destructive — it consumes 003's fresh state, the exact state class Journey 6 needs.
- Journey 9 (CAPA): `CAPA-2024-002`'s Verification tab and a MAJOR CAPA's Approval tab both load.

☐ **Reseed after spot-checking.** The Journey 6 spot-check failed part 003, so run `seed_demo` once more and log back in as **Sarah** — the class must start with 003 fresh. (Do this last; the reseed also invalidates the logged-in sessions.)

☐ Read *What to skip in inspector training* at the bottom so you can defer out-of-scope questions cleanly.

Roles you'll play

All demo logins use password **`demo123`**.

Login	Name	Role	You play them in
`sarah.qa@demo.ambac.com`	Sarah Chen	QA Inspector	Everything — the trainee's identity
`maria.qa@demo.ambac.com`	Maria Santos	QA Manager	The co-sign in Journey 6; the manager block
`mike.ops@demo.ambac.com`	Mike Rodriguez	Operator	Not logged into — his signatures are pre-seeded on part 001's first-piece substeps

At a glance — the arc and its pacing

Eleven inspector journeys, ~50 min; the manager block is separate and optional. Rough budget:

#	Journey	Min	The one thing they leave with
1	Read the home page	5	which chip is today's queue
2	The scanner is your GPS	3	a scan anchors on the parent WO
3	Receiving inspection	6	accept a lot through the DWI
4	FPI buy-off	6	the buy-off is a batch gate (and SoD)
5	Sampled inspection	5	"why this part?" = the Sample flag

#	Journey	Min	The one thing they leave with
6	Live fail + disposition	9	derived FAIL → quarantine → co-signed decision
7	Re-inspection	4	read the history before re-inspecting
8	OSP return inspection	4	inspect it before it rejoins the flow
9	CAPA task	5	disposition (this part) vs CAPA (the pattern)
10	Calibration & notifications	3	check the gauge's cal before use
11	Reading finished records	4	narrate a record backward

If you're short on time, **Journey 6 is the one to protect** — it's the core of the job and the only gate that needs a second person.

What you'll teach against

One demo work order, **WO-QA-INSPECT-01** (order *QA Inspector Onboarding Walkthrough* / `ORD-2024-QA-INSPECT`, customer **Midwest Fleet Services**, Common Rail Injector, Injector Reman process). Each part is parked at the exact state its journey needs; a reseed restores all of it. (Full detail: `WO-QA-INSPECT-01_reference.md`.)

Serial	Journey	State the trainee sees
`INJ-QA-INSPECT-001`	4 — FPI buy-off	IN_PROGRESS @ Nozzle Inspection, first piece complete, **PENDING FPI**
`INJ-QA-INSPECT-002`	5 — Sampled inspection	AWAITING_QA @ Nozzle Inspection, **Sample** ("Post-repair verification")
`INJ-QA-INSPECT-003`	6 — Fail + disposition	IN_PROGRESS @ Flow Testing, fresh — you fail it **live**
`INJ-QA-INSPECT-004`	7 — Re-inspection	AWAITING_QA @ Flow Testing, visit 2 (historical FAIL QR + CLOSED REWORK disposition)
`INJ-QA-INSPECT-005`	8 — OSP return	AWAITING_QA @ Nitride Coating, RETURNED from Apex Plating
`INJ-QA-INSPECT-006`	1 (background)	QUARANTINED @ Assembly, bare OPEN disposition on Sarah

Two states WO-QA-INSPECT-01 doesn't stage — a part mid-rework and a scrapped closed record — are borrowed from the older demo storyline, exactly as the walkthrough's §12b does:

Serial	Journey	State
INJ-0042-019	11 — Rework in flight	REWORK_IN_PROGRESS, disposition IN_PROGRESS
INJ-0042-023	11 — Closed record	SCRAPPED, CLOSED SCRAP disposition + FAIL QR trail

Journey 1 — Log in and read the home page (walkthrough §1)

Why: Everything starts here. Before touching a scanner, the trainee should recognize each block on the home page and know what it's telling them.

Setup: you're already logged in as Sarah, on `/` (the QA home; it also answers at `/quality/inbox`).

You see, top to bottom:

- **Scan box** — one field; anything typed or scanned resolves to a work order and drops you on WO Detail.
- **FPI banner** (red border, siren icon) — one row per pending First Piece Inspection: step, work order, part, and a waiting-age timer. Buttons **I'm on it** (acknowledge) and **Start check** (jump to the WO's Control page). Two rows here: the walk's own `INJ-QA-INSPECT-001` and a pre-existing `WO-2024-0048-A` row.
- **Inbox chips** — filter the flat list by type: **All · Receiving · OSP returns · In-process**. Each carries a count and the age of its oldest item.
- **Urgent** — a red count pill after the chips. It's plain text, *not* a filter — don't hunt for a click target.
- **My quality actions** — three tiles (Approvals, CAPA tasks, My dispositions) counting work assigned to Sarah; the "N overdue" badge is the signal.
- **Your gauges** — calibration status on gauges Sarah used recently (*Torque Wrench TW-25 — overdue*). A compliance cue.

Watch for:

- Trainee reads the Inbox count but ignores the chips. Read each chip aloud — the shape of the day depends on which one is full.
- Trainee ignores *Your gauges*. That's the compliance signal.

Checkpoint: trainee can name each block and say what it's telling them.

Journey 2 — The scanner is your GPS (walkthrough §1)

Why: The barcode reader isn't a data-entry tool; it's a navigator. Every scan resolves to the **parent work order** and lands you on WO Detail — the shared surface where exceptions live and the DWI capture path begins.

You do: scan the **header barcode** on the WO-QA-INSPECT-01 traveler (or type `WO-QA-INSPECT-01` in the scan box).

You land on: `/workorder/$workOrderId` — **WO Detail** (same landing for QA and operators).

You see: heading `WO-QA-INSPECT-01` with status/priority; an action bar (**Traveler**, **Start Work**, **Hold**, **Cancel**); and **Overview** / **Parts** tabs. Overview carries the exception badges (AWAITING QA / QUARANTINED / REWORK NEEDED), the OSP card, and the digital traveler; Parts is the per-part drill-down.

Then: type or scan any `INJ-QA-INSPECT-###` part number in the scan box (its label reads *"Scan or type a work order / part number"*) — you land on the **same** WO Detail (part → parent WO). For the *part* detail (`/details/Parts/$id`), open it from the Parts-tab row.

Teaching point: the WO is the anchor for scans; per-part detail is one click away. The **Control** page (`/workorder/$id/control`) is a *different* lens — lead/manager oversight, per-step status, OSP actions — not a scan destination.

Checkpoint: trainee scans the traveler and, unprompted, points at the exception badges and says what each filters.

Journey 3 — Receiving inspection (walkthrough §2)

Why: Material comes in the door before it's ever a WO part. Receiving is where a lot is accepted into stock or held.

You do: from the home Inbox, click the **Receiving** chip (or go to `/production/receiving-inspection`).

You see: the **Receiving Inspection Queue** — *Lot # · Material · Supplier · Qty · Status · Actions*, purchased lots only (5 rows on a fresh seed), each with **Inspect**. `RCV-INJ-HOLD` (Bargain Bolts, QUARANTINE, "Unqualified supplier") is a hold exhibit — don't inspect it.

You do: Inspect `RCV-INJ-0001` (Great Lakes, 250 EA) → the lot detail shows the CoC panel, the **Sample plan (C0, level III, TIGHTENED): Inspect 29 of 250 · Accept ≤ 0 · Reject ≥ 1**, and **Run Inspection (DWI)**.

You do: Run Inspection (DWI) → the operator runtime. Enter a passing **Outer Diameter** (`25.01`), set the result **Pass**, sign, **Confirm & review**, then **Accept lot**. Toast *"Lot accepted"*; the lot leaves AWAITING_INSPECTION.

Teaching point: receiving uses **Accept lot / Reject lot** (not "Complete step") — the domain verb. A **Fail** would open the reject-disposition dialog; that's the supplier-quality path, out of scope for a new inspector.

Watch for: the footer names missing required fields but never blocks — **Confirm & review** stays enabled and just scrolls you to the first gap.

Checkpoint: trainee can state the difference between the receiving-only queue and the unified `/production/incoming` queue (purchased-only vs. purchased + OSP returns).

Journey 4 — First Piece Inspection buy-off (walkthrough §3)

Why: On a step's first pass, one part is the **first piece**. The operator runs the DWI on it; QA buys it off before the rest of the batch runs. A failed or missing FPI blocks the whole batch — it's a production gate.

You do: on the home FPI banner, find the `INJ-QA-INSPECT-001 · Nozzle Inspection` row. Click **I'm on it** (it reads *"Seen by Sarah"*).

You reach the runtime: WO Detail → **Start Work** → tick `INJ-QA-INSPECT-001` under Nozzle Inspection → **Start**. The runtime opens with the inspection substeps **already signed by Mike** (the operator) and reads **"2 of 2 confirmed"** — the first piece is complete and waiting on QA.

You see the FPI banner with three buttons: **Sign off & pass · Fail · Waive**.

You do: Sign off & pass → confirmation dialog (*"By signing off you attest… conforms. This is recorded against your name and releases the run."*), add a note, **Confirm sign-off**. The FPI flips to **PASSED**, recorded against Sarah, and the banner turns green — *"Setup verified — all parts can proceed."*

Teaching point (segregation of duties): the substeps were signed by Mike; Sarah signs the buy-off. Different people — that's the SoD rule. If Sarah had also signed the substeps, the buy-off would refuse.

Manager's side: an inspector who *lacks* `sign_off_fpi` isn't stuck — the banner offers an inline **co-signature** for an authorized colleague. (Sarah has it, so she signs directly.)

Checkpoint: trainee explains why the FPI is a batch gate, not a per-part check.

Journey 5 — An inspection sampled to you (walkthrough §4)

Why: Sampling triage is the most common daily task — a rule flags a part, production parks it, it lands in your Inbox, you clear it.

Order matters: do Journey 4 first. The Nozzle Inspection FPI gates the whole step — on a fresh seed, if 001's FPI isn't signed off yet, 002's runtime shows *"First Piece Inspection in progress"* and blocks (walkthrough §4c).

You do: home Inbox → **In-process** chip → the `Nozzle Inspection · 1 pcs · WO-QA-INSPECT-01` row. That's `INJ-QA-INSPECT-002`.

You see: it lands on the WO **Control** page; in the Step Status table, `INJ-QA-INSPECT-002` is `AWAITING_QA` at Nozzle Inspection with a **Sample** flag. Its sampling reason is *"Post-repair verification"* — it had earlier rework, so the rule pulls it for extra scrutiny. Open the part detail (`/details/Parts/$id`) to show **Rework Passes 1**, **Sampling Required Yes**.

You do: run the inspection from **WO Detail → Start Work → tick 002 → Start** (the Control table's rows route steps, they don't launch the runtime). Work the Nozzle Inspection DWI — the visual points and the measurement — sign, **Confirm & next → Complete step**. The part advances.

Watch for:

- Trainee submits a measurement without confirming the gauge and its cal state. Habit-build: "Which gauge? Is it in cal?" before every capture.
- The classic new-inspector error — conflating "measurement in spec" with "the part passed." One in-tolerance reading is one characteristic; the verdict is the whole inspection. If they blur the two, walk it again.

Checkpoint: trainee can answer "why this part?" by pointing at the Sample flag and the sampling reason.

Journey 6 — A live fail and its disposition (walkthrough §5 → §6)

Why: The core of the job — a reading is out of spec, the system quarantines the part, and someone has to decide what to do with it. This journey has the one gate a lone inspector can't clear alone, so keep Maria's login ready.

6a — Fail it live

You do: WO Detail → **Start Work** → tick `INJ-QA-INSPECT-003` (Flow Testing) → **Start**. (Flow Testing carries an FPI gate too, but the seed pre-signs it, so 003 opens without an FPI block — you'll see a green "First Piece Inspection signed off" banner.) On the runtime, enter an **out-of-spec Flow Rate** — `98` (below the 100 LSL) — scan the part, tick the flow-bench-in-cal attestation.

Teaching point: there's **no Pass/Fail button** here — the verdict is *derived* from the measurement crossing its spec. The spec is the judge.

You do: Confirm & review → Complete step.

What this triggers (walk it on screen): the part goes **QUARANTINED**, a **FAIL Quality Report** is written, and a **disposition is auto-created** (OPEN, no type). Sarah's *My dispositions* count ticks up.

6b — Work the disposition (needs a co-sign)

You do: open the disposition (part detail → **Dispositions** widget → edit, or `/production/dispositions` → find `INJ-QA-INSPECT-003`). In the editor set **Disposition Type = REWORK**, **Severity = MAJOR**, a containment action and resolution note, then **Update Disposition**.

What happens: because Sarah is a QA Inspector and lacks `approve_disposition`, the editor **does not commit** — it opens the **Authorize disposition decision** co-sign dialog. Choosing a disposition type is the authorized act (AS9100 8.7).

You do: enter **Maria's** email (`maria.qa@demo.ambac.com`), draw the signature, tick *"I authorize this disposition decision,"* enter her password (`demo123`), **Authorize**. The decision commits **recorded under Maria**, the disposition goes IN_PROGRESS, and the part moves to REWORK_NEEDED (rework count +1).

Teaching point: this is a *co-signature*, not a login switch — Maria never logs in; she authenticates on Sarah's screen and the record carries her name. A QA Manager working it directly commits without the dialog.

Watch for: trainee tries to finish on Sarah's login alone and reads the co-sign dialog as an error. It isn't — it's the gate.

Checkpoint: trainee can say who the decision is recorded against, and why an inspector co-signs rather than being blocked.

6c — Closing the disposition (this one's yours)

Note the split before moving on: the *decision* needed a co-sign, but **closing** the disposition is the inspector's own — Sarah holds `close_disposition`. In the live flow you'd close it once the rework is done and re-inspected (Journey 7): from the editor set **Current State = CLOSED** (or the **Close** action). Closing is refused unless the completion blockers are clear — containment recorded for MAJOR/CRITICAL, a decision selected, no pending 3D annotations — and a rejected close still saves your other edits, so you fix the blocker and retry (§6d).

Teaching point: authoring the *decision* (co-sign) and *closing* the record (inspector) are two different permissions on the same disposition.

Journey 7 — Re-inspecting a reworked part (walkthrough §7)

Why: After rework, the part comes back to QA for a second look. Reading the history *before* re-inspecting is the discipline.

You do: home Inbox → the `INJ-QA-INSPECT-004 · Flow Testing` row. Open the part detail first: it carries the trail — an original **FAIL QR** (flow rate 98) and a **CLOSED REWORK** disposition (`DISP-QAI-004-REW`), and it's **visit 2**.

You do: run the re-inspection from WO Detail → Start Work → 004 → Start. Enter a passing value this time; complete the step. A second **PASS** QR is written and the paper trail reads end to end: FAIL → CLOSED REWORK → reworked → PASS.

Checkpoint: trainee reads the part's history and can narrate why it's back.

Journey 8 — OSP return inspection (walkthrough §8)

Why: Some steps are done by a subcontractor. Parts ship out, come back, and must be re-inspected before rejoining the routing.

You do: home Inbox → **OSP returns** chip → the Apex Plating return for `INJ-QA-INSPECT-005` (shipment `OSP-2026-####` — the number rotates; it's the Nitride Coating return). Or reach it from `/production/incoming` (the unified queue, which shows a *Source* column).

You do: open the return inspection and work it like any DWI capture.

Teaching point: the receiving-only queue is purchased material; the unified **Incoming Inspection** queue adds OSP returns — same runtime, wider net.

Checkpoint: trainee can say why an OSP return needs inspection before it rejoins the flow.

Journey 9 — Working a CAPA task (walkthrough §9)

Why: When a failure is systemic (a pattern, not a one-off part), it becomes a CAPA — a structured investigation. Inspectors work assigned CAPA tasks and can initiate one; the effectiveness verification and management approval are the QA Manager's.

You do: home → *My quality actions* → *CAPA tasks* (or **CAPAs** in the rail → `/quality/capas`). Open **CAPA-2024-002** (Pending Verification, 75%).

- **Verification tab** — *"Effectiveness Verification."* Sarah can author the verification **plan** (Add Verification → method + success criteria). Recording the **outcome** (Complete Verification → CONFIRMED / NOT_EFFECTIVE) needs `verify_capa` — a QA Manager, co-signable like the disposition.
- **Approval tab** — a MAJOR/CRITICAL CAPA shows *"Awaiting Approval — work cannot begin until approved."* Read-only for Sarah; approving is the QA Manager (Journey M below).
- **Initiate** — from a failed QR's **Create CAPA**, when a pattern warrants it. Don't open a CAPA for every fail — the disposition already handles *this* part.

Checkpoint: trainee can say the difference between a disposition (this part) and a CAPA (the pattern).

Journey 10 — Calibration and notifications (walkthrough §10–§11)

Why: The ambient signals — is my gauge in cal, and what fired for me.

- **Calibration** (`/quality/calibrations`): the **Calibration Dashboard** (Equipment / Current / Due Soon / Overdue / Compliance). The home *Your gauges* tile is the point-of-use nag; a gauge overdue for cal shouldn't be used.
- **Notifications** (bell popover / `/notifications`): the feed of what fired for Sarah — CAPA assignments, `ncr.opened` on a fail, etc. Show *Mark all read* and the filters.

Checkpoint: trainee checks a gauge's cal state before using it, unprompted.

Journey 11 — Reading finished records (walkthrough §12)

Why: Half the job is *reading* records others created — a part mid-rework, a scrapped part — and reconstructing the story.

- **Rework in flight:** open `INJ-0042-019` (part detail) — REWORK_IN_PROGRESS, a linked FAIL QR and an **IN_PROGRESS** disposition. The story is "failed, dispositioned to rework, currently being reworked."
- **Closed scrap record:** open `INJ-0042-023` — SCRAPPED, with a FAIL QR (porosity / failed hold pressure) and a **CLOSED SCRAP** disposition carrying scrap verification. The story reads backward from the terminal state.

(These two live on the older demo WO-2024-0042-A because WO-QA-INSPECT-01 doesn't stage a mid-rework or scrapped part — the walkthrough borrows the same `INJ-0042-023` for its §12b.)

Teaching point: a part with no linked QR or disposition behind a QUARANTINED/SCRAPPED status is a red flag — how did it get there without a record?

Checkpoint: trainee opens a closed record and narrates the sequence that produced it.

Optional manager block (walkthrough §13)

If a lead or QA Manager is in the room, show the *other side* of the gates the inspector journeys handed up — log in as **Maria** (`maria.qa@demo.ambac.com`):

- **`/approvals`** — the approvals center: *Awaiting My Approval*, *By Type*, *My Submitted Requests*.
- **Authorize a disposition** directly (holds `approve_disposition` — no co-sign dialog).
- **Verify a CAPA's effectiveness** on `CAPA-2024-002` (Complete Verification → CONFIRMED / NOT_EFFECTIVE). Note the dependency: the verification *plan* must already exist (Journey 9 authors it) — on a truly fresh seed there's no plan yet, so no **Complete Verification** button appears until one is added (§9c/§13c).
- **Approve a MAJOR CAPA** — `CAPA-2024-005`, which sits in her queue; Submit Response → Approve with signature + password (§13d).
- The co-sign-vs-queue distinction (§13e): *is someone blocked at a gate right now (co-sign), or did a request land in my queue to action later (approval)?*

What to skip in inspector training

Defer these cleanly — they're real but not a new inspector's job:

- **Process / DWI authoring** (walkthrough §14) — building processes, measurements, sampling rules, and substeps is a QA-manager / process-author task. If asked, point to §14 and move on.
- **Change control, supplier quality, part approvals** — adjacent QMS surfaces, not day-one inspection.
- **The manage-all-records editors** (`/editor/qualityReports`, etc.) — an inspector's QRs reach them through the part and disposition (§5/§6), not by trawling the register. The register itself is §15 reading, not a task.
- **Exact record numbers** — `DISP-2026-####` / `QR-2026-####` / `OSP-2026-####` rotate on reseed. Never make a trainee memorize one.

Refreshing screenshots

This script was rebuilt onto the WO-QA-INSPECT-01 scenario; its screenshots have not yet been recaptured and the inline image references were removed rather than ship stale ones. To add them:

1. Start the dev servers; fresh `seed_demo`; log in as **Sarah**.
2. Capture at **1440×900** — the training standard.
3. One tight shot per journey (crop out browser chrome).

4. Save under `Documents/screenshots/qa_inspector_training/` and add the `![alt](…)` reference at the end of each journey.
5. The demo tenant is fictional, but glance for accidental real names before committing.
6. Rule of thumb: if a screenshot ever stops matching the live screen, the screenshot is wrong, not the software — recapture it.